

Blood Cells and Hematopoiesis

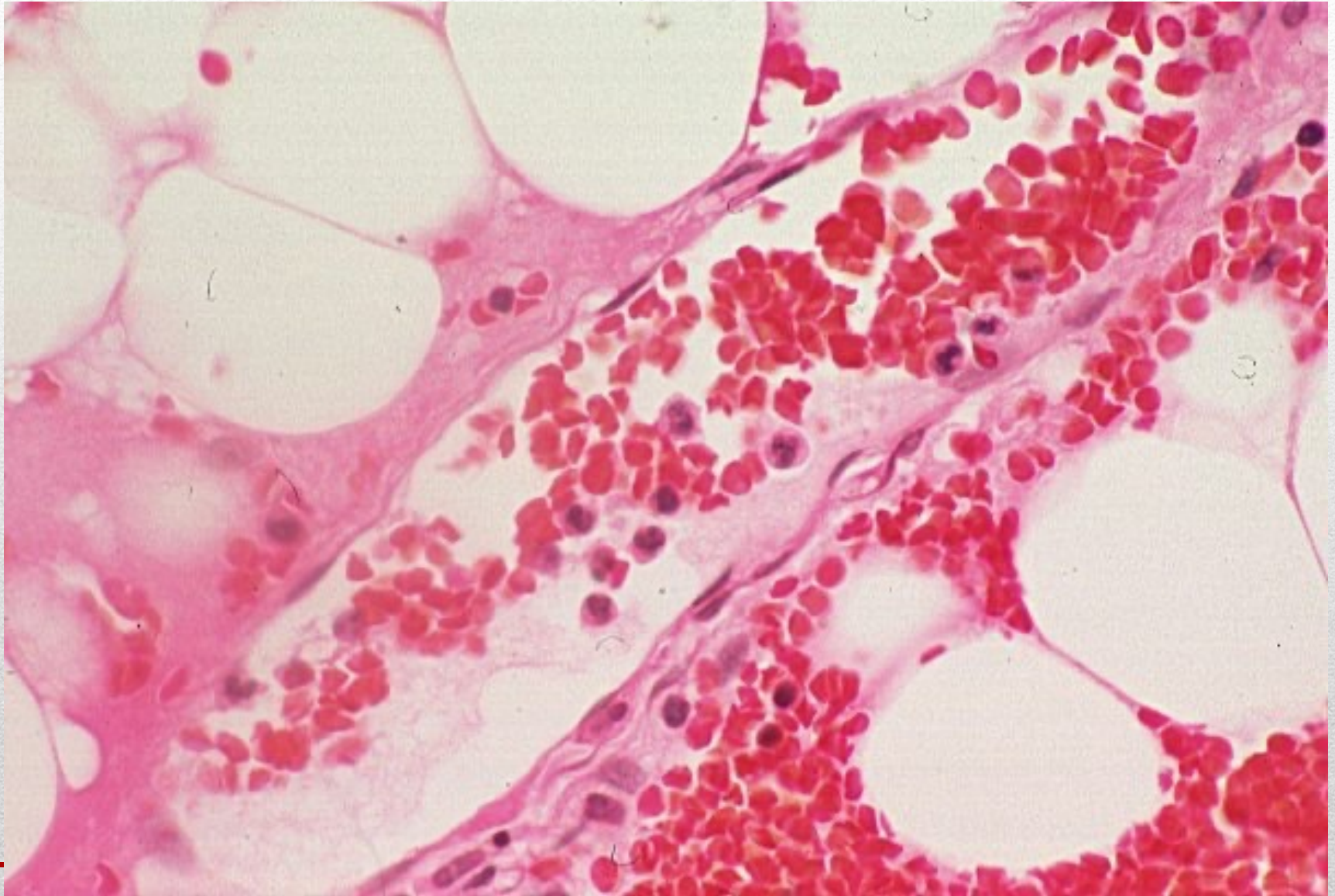


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Formed Elements of the Blood

- Red Blood Cells (Erythrocytes)
 - Platelets
 - White Blood Cells
 - Granulocytes
 - Neutrophil
 - Eosinophil
 - Basophil
 - Agranulocytes
 - Monocytes
 - Lymphocytes (small and large)
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Erythrocytes (RBCs)



Erythrocytes (RBCs)

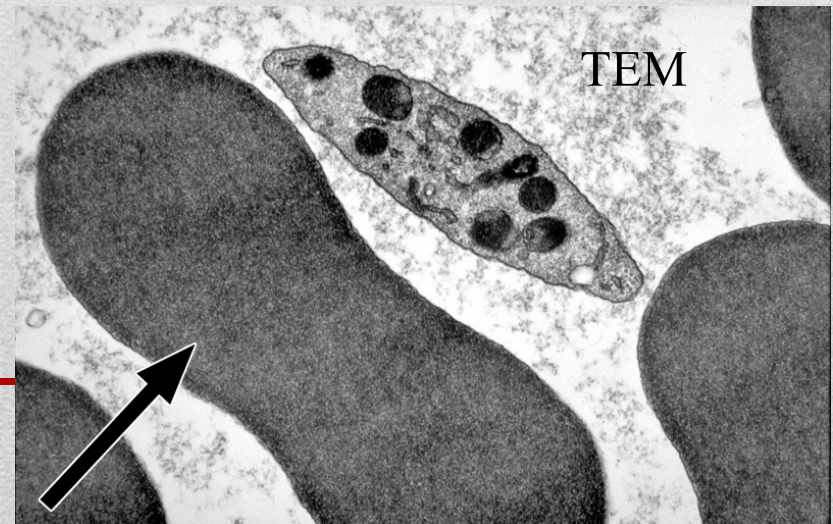
Anucleate, non-granulated, eosinophilic cells. Uniform in shape (biconcave discs) and size (7.2 microns).

Central concavity that increases surface area of cell for gas exchange. The concavity appears pale under the light microscope.

These cells contain hemoglobin and are responsible for the transport and delivery of oxygen.

Erythrocytes have a lifespan of 120 days.

By EM – a homogeneous cytoplasm



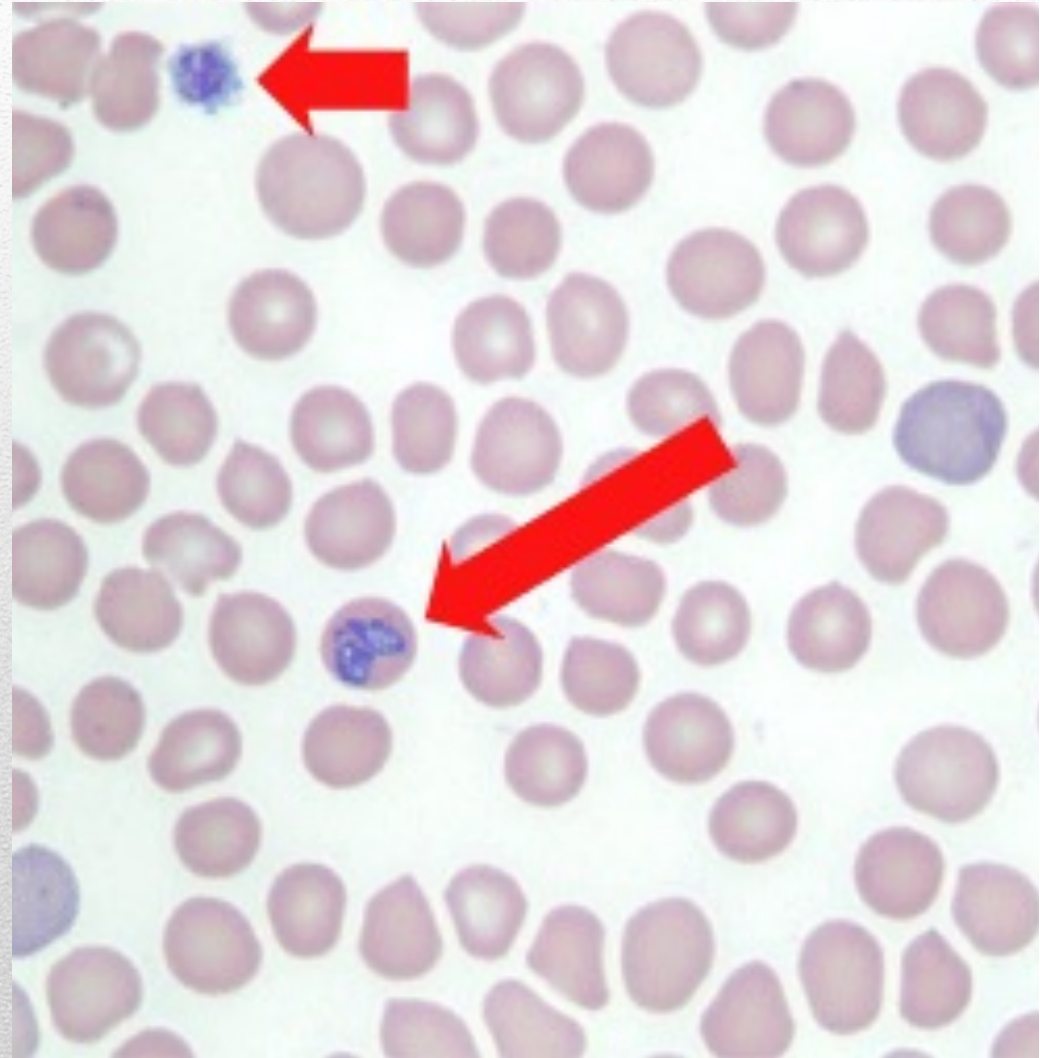
Cells that extrude their nucleus and fill their cytoplasm with one class of proteins in the mature state.

- Red blood cells extrude their nucleus and fill their cytoplasm with hemoglobin
 - Keratinocytes in the skin extrude their nuclei and fill their cytoplasm with types of keratins.
 - Lens fibers extrude their nuclei and fill their cytoplasm with crystallins.
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Reticulocytes

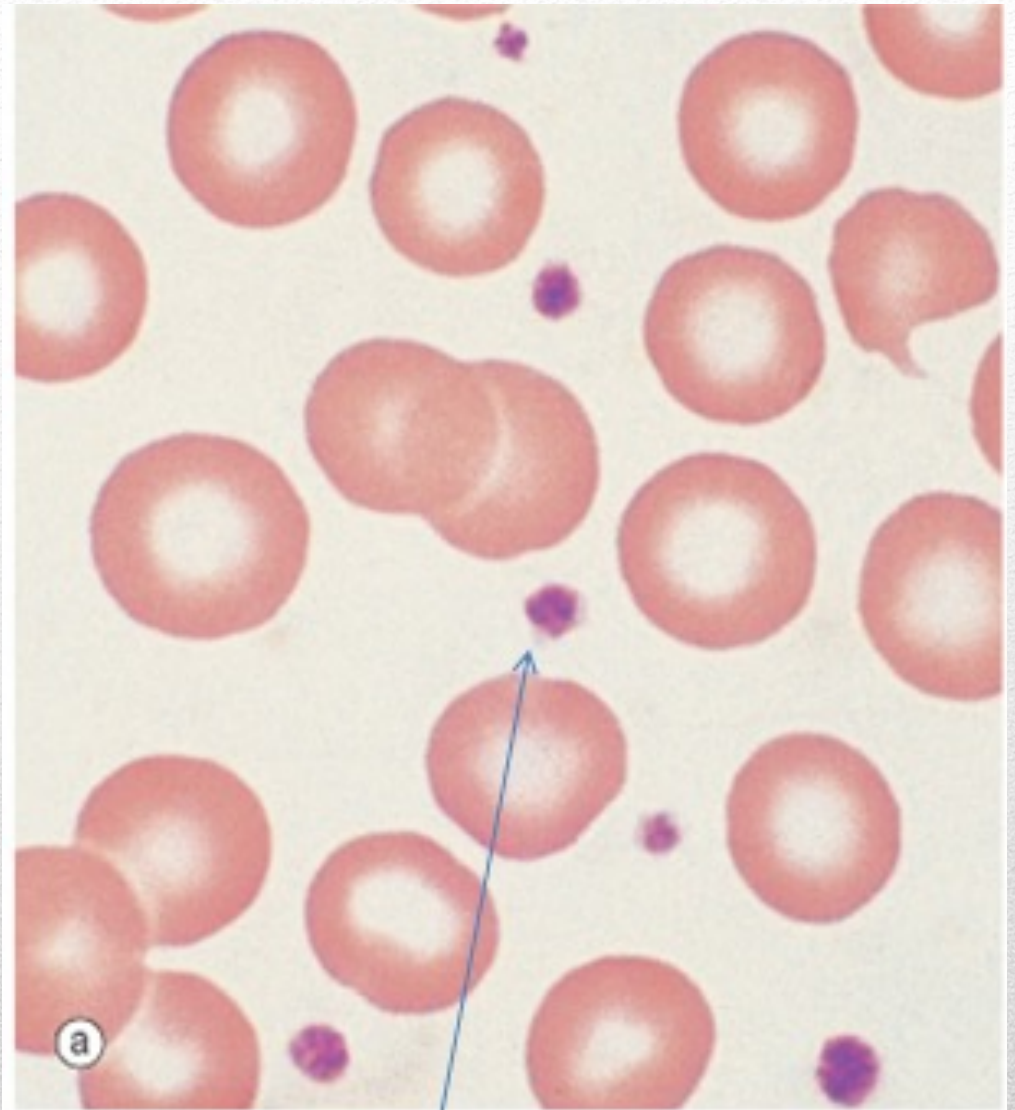
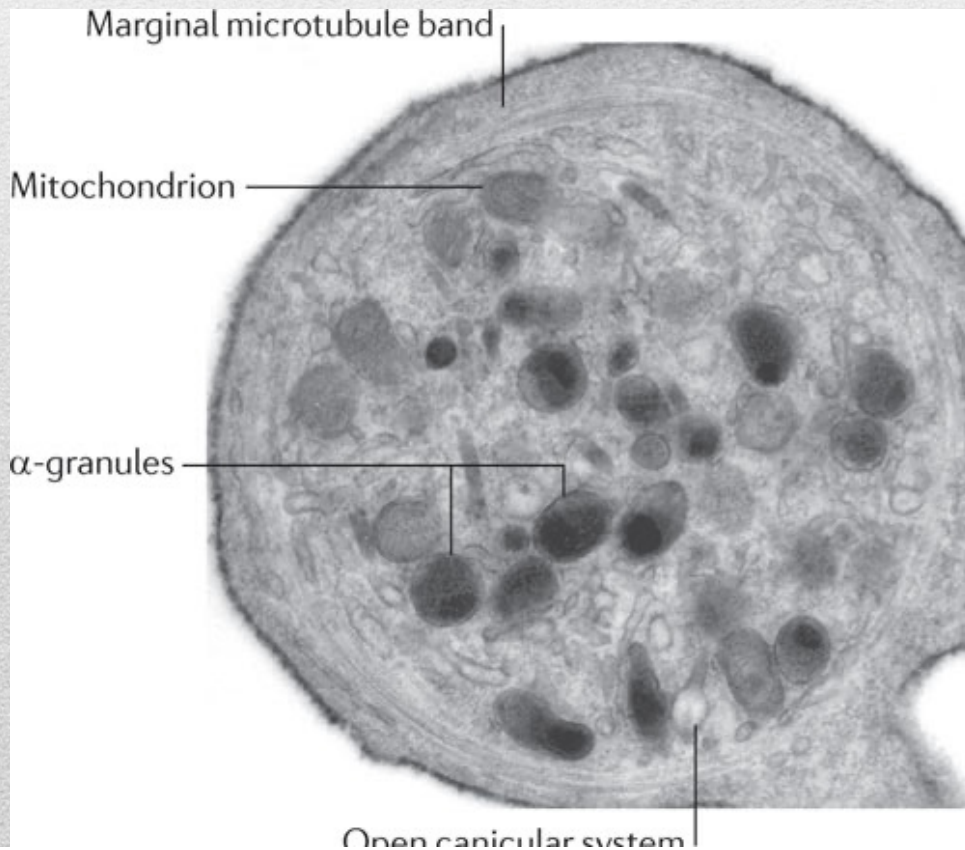
Reticulocytes are immature red blood cells that are prematurely released from the bone marrow. They mature into erythrocytes after 1 to 2 days in the peripheral blood.

In a normal blood smear it is normal to find about 1 reticulocyte for every 100 red blood cells. These cells exhibit strands and dots of blue within them because they still have RNA-containing organelles like free ribosomes.



Platelets

Microtubule band maintains sphericity and produces contraction when activated. *α -granules* contain proteoglycans, hemostasis factors and cofactors. *δ -granules or dense bodies* store serotonin and Ca^{++} , *λ -granules* are lysosomes,



2-3 microns in size, 150,000 to 450,000 per ml
Lifetime 5-11 days

Platelets

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Preparation of blood smear

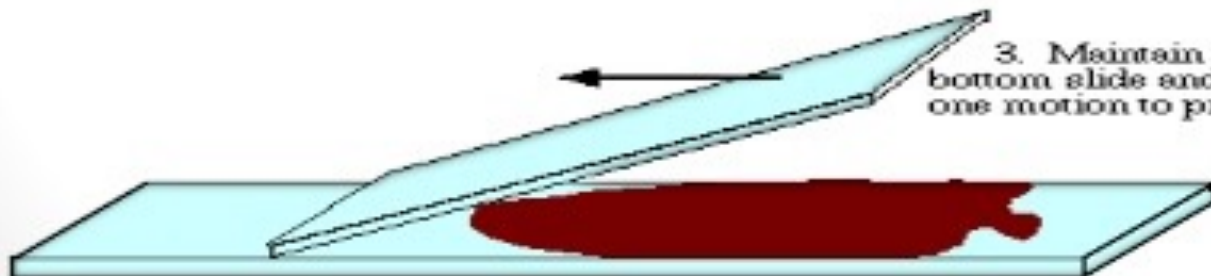
1. Place a small drop of whole blood on a **VERY CLEAN** slide. Hold a second slide at the angle shown.



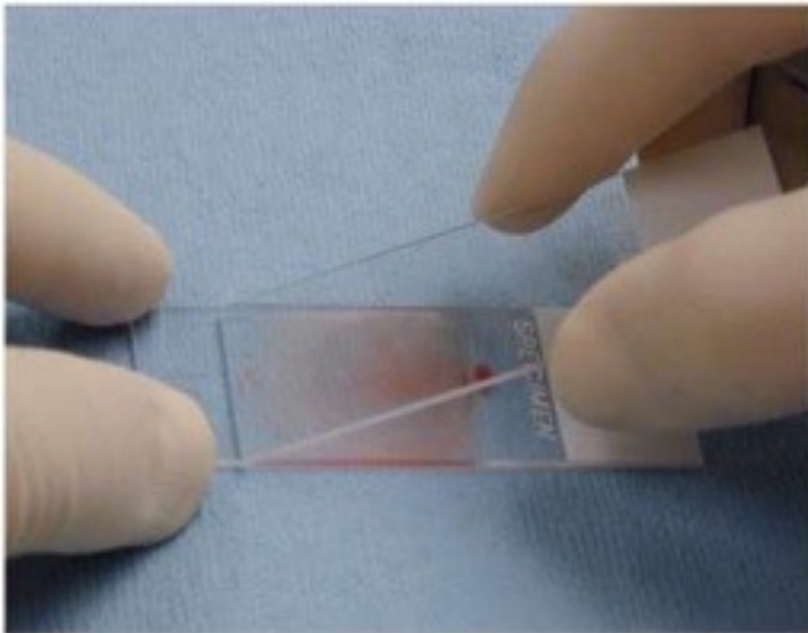
2. While maintaining contact with the bottom slide, pull the top slide back to contact the drop, which will spread by capillary action.



3. Maintain firm contact with the bottom slide and push the top slide in one motion to produce the smear.



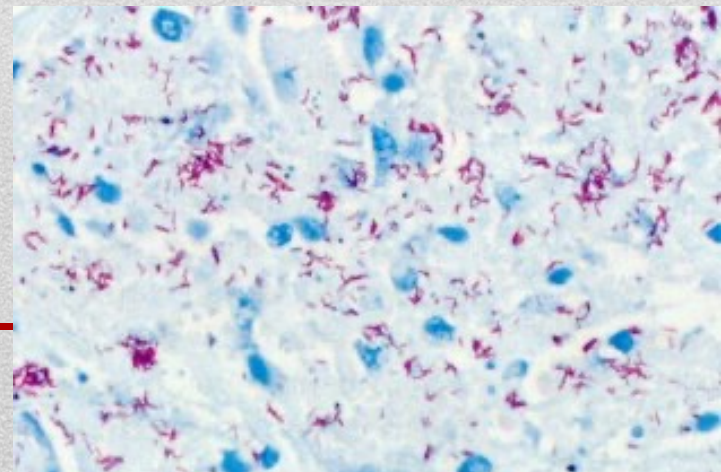
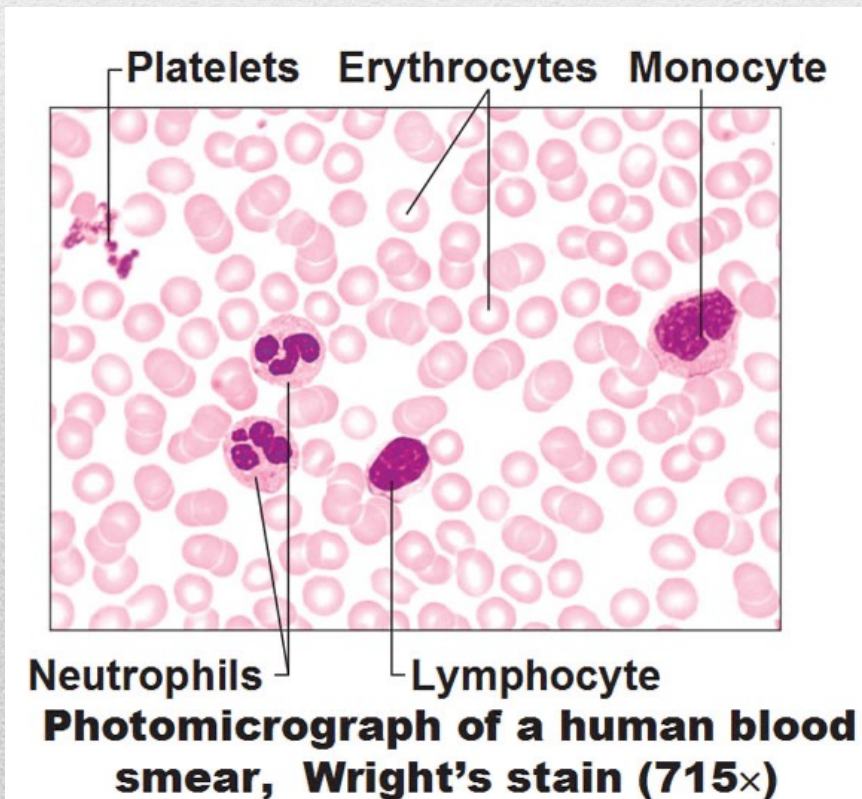
The shape of blood film



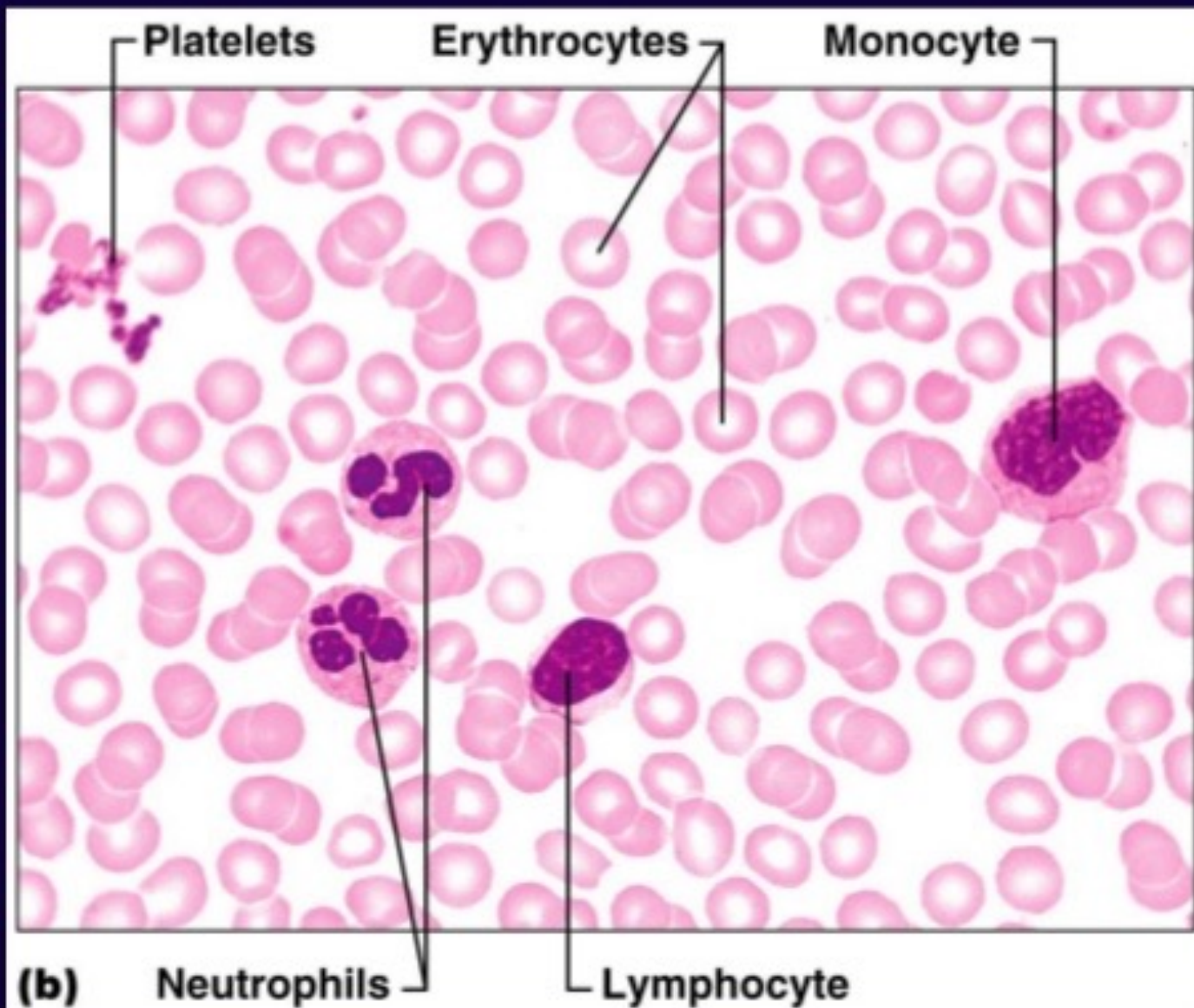
Stains for Blood Smears –

Most common are Wright stain and Giemsa stain

- **Wright Stain** – works for routine blood smears
 - **Eosin** - stains cytoplasm pink
 - **Methylene blue** – stains nuclear material/nucleic acids blue
- **Giemsa Stain** – preferred if systemic microbial disease suspected.
 - Eosin – stains cytoplasm pink
 - Methylene blue – stains nuclear material/nucleic acids blue
 - Azure B – especially good at identifying reticulocytes
- Erythrocytes stain pink, platelets show a light pale pink, lymphocyte cytoplasm stains sky blue, monocyte cytoplasm stains pale blue, and leukocyte nuclear chromatin stains magenta



Wright's Stain



White Blood Cells

Granulocytes

Neutrophil

Eosinophil

Basophil

Agranulocytes

Monocytes

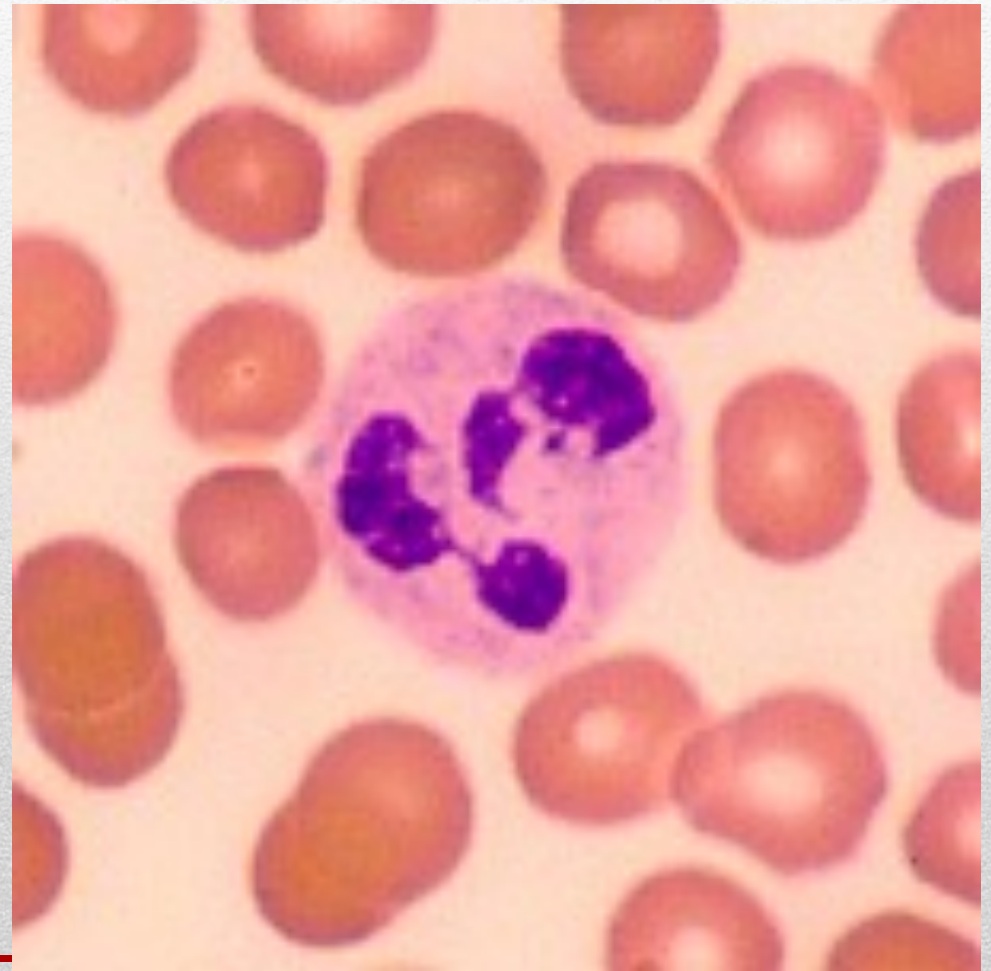
Lymphocytes (small and large)

White Blood Cells

Type Of White Blood Cells	% By Volume Of WBC	Description	Function
Neutrophils	60 – 70 %	Nucleus has many interconnected lobes; blue granules	Phagocytize and destroy bacteria; most numerous WBC
Eosinophils	2 – 4 %	Nucleus has bilobed nuclei; red or yellow granules containing digestive enzymes	Play a role in ending allergic reactions
Basophils	< 1 %	Bilobed nuclei hidden by large purple granules full of chemical mediators of inflammation	Function in inflammation mediation, similar in function to mast cells
Lymphocytes (B Cells and T Cells)	20 – 25 %	Dense, purple staining, round nucleus; little cytoplasm	the most important cells of the immune system; effective in fighting infectious organisms; act against a specific foreign molecule (antigen)
Monocytes	4 – 8 %	Largest leukocyte; kidney shaped nucleus	Transform into macrophages; phagocytic cells

Granulocytes

- **The Neutrophil:**
- This granulocyte has very tiny light staining granules (the granules are very difficult to see because they are neutrophilic and do not stain with blood smear stains). The nucleus is frequently multi-lobed with lobes connected by thin strands of nuclear material. These cells are capable of phagocytizing foreign cells, toxins, and viruses.
- When taking a Differential WBC Count of normal blood, this type of cell would be the most numerous. Normally, neutrophils account for 50-70% of all leukocytes.



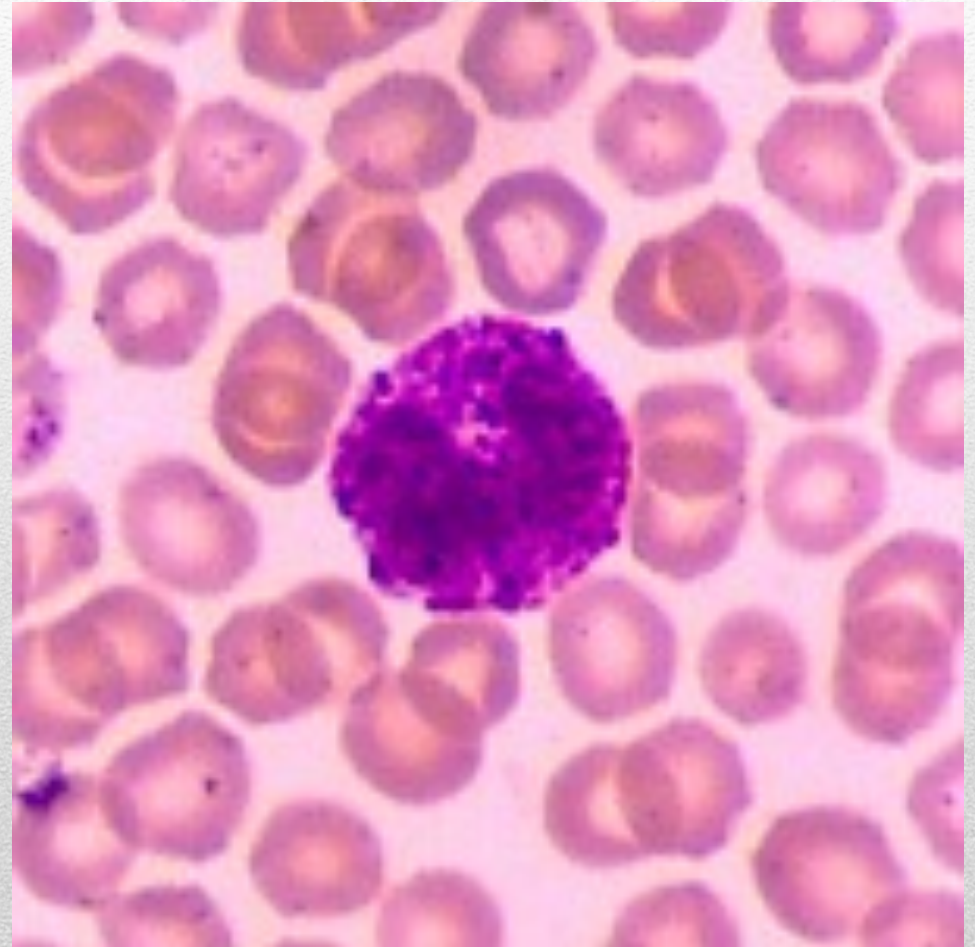
Granulocytes

- “Bands”
- Band neutrophils are an intermediary step prior to the complete maturation of segmented neutrophils. Horseshoe shaped nucleus not yet fully segmented. An increase in band neutrophils typically means that the bone marrow has been signaled to release more WBCs and increase production of WBCs, also known as a "left shift". Most often this is due to infection or inflammation in the body.
- Blood reference ranges for neutrophilic band cells in adults are 3 to 5% of white blood cells,^[3] or up to $0.7 \times 10^9/L$.



Granulocytes

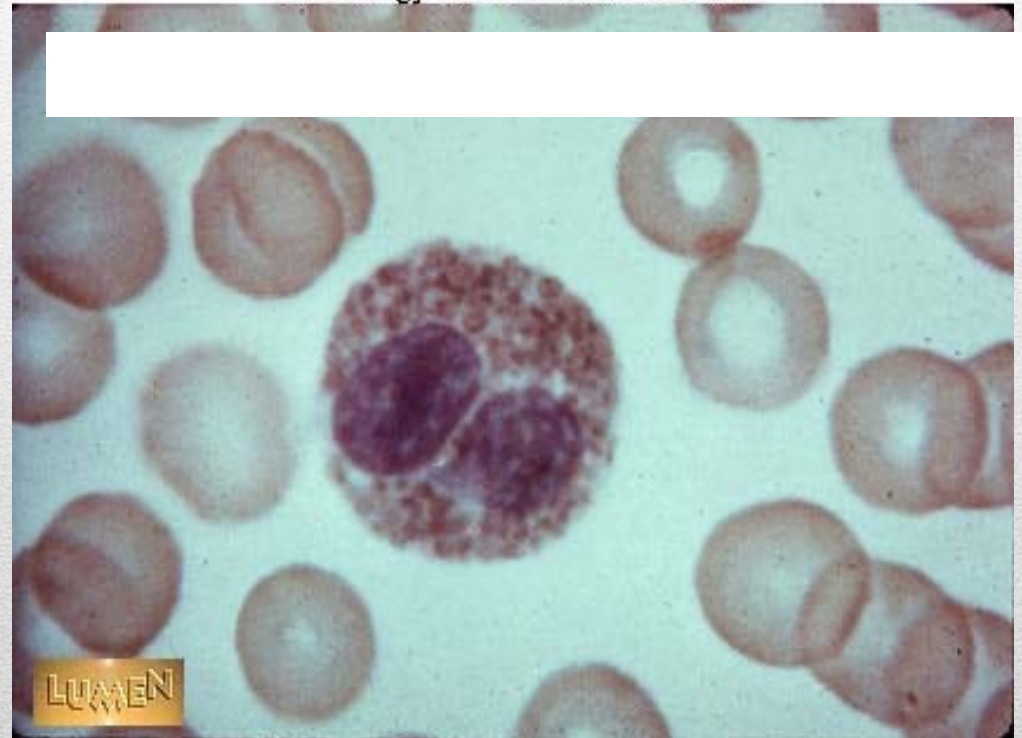
- **The Basophil:**
- The basophilic granules in this cell are large, stain deep blue to purple, and are often so numerous they mask the nucleus. These granules contain histamines (cause vasodilation) and heparin (anticoagulant).
- In a Differential WBC Count we rarely see these as they represent less than 1% of all leukocytes.



Granulocytes

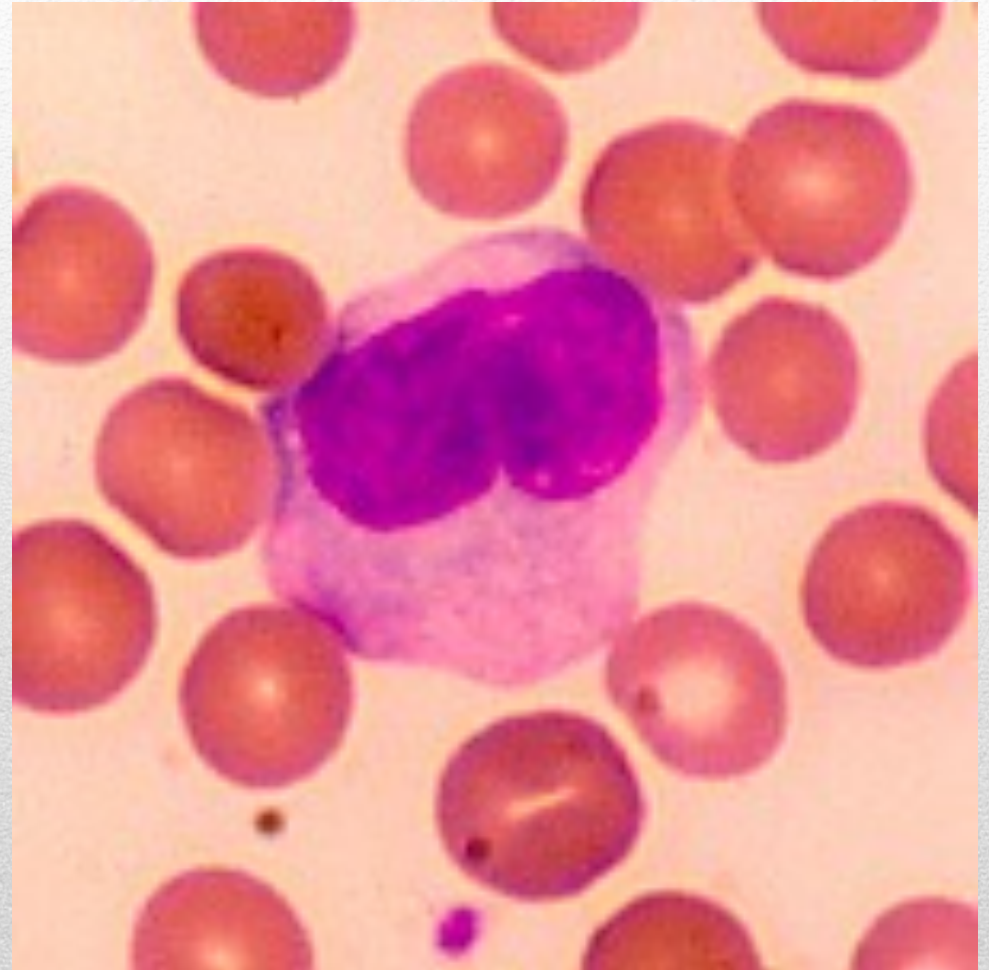
- **The Eosinophil:**
 - A granulocyte has large granules that are pink (or red) in a stained preparation. The nucleus is bi-lobed. The granules contain digestive enzymes that are particularly effective against parasitic worms in their larval form. These cells also phagocytize antigen - antibody complexes.
 - These cells account for less than 5% of the WBC's.
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Histology Lab Part 5: Slide 52



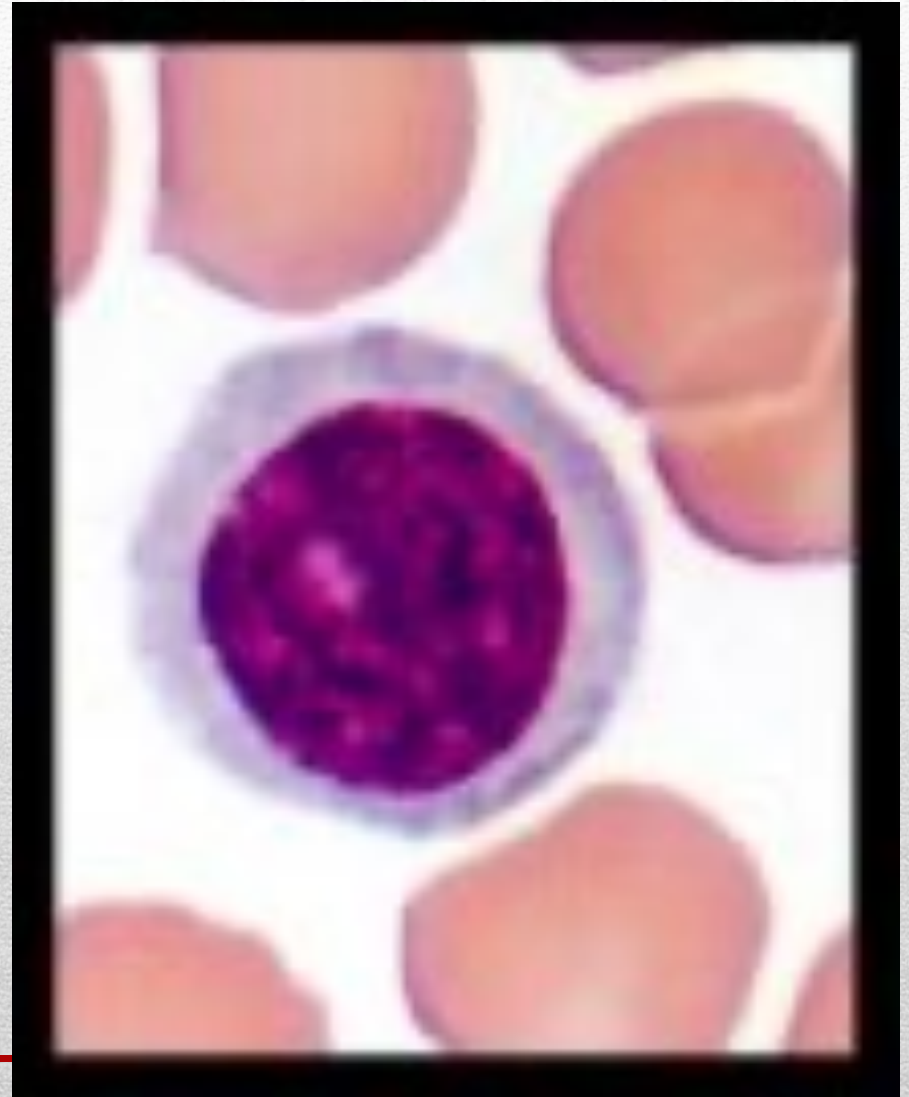
Agranulocytes

- **The Monocyte:**
- This cell is the largest of the leukocytes and is agranular. The nucleus is most often "U" or kidney bean shaped; the cytoplasm is abundant and light blue (more blue than this micrograph illustrates). These cells leave the blood stream (diapedesis) to become macrophages. As a monocyte or macrophage, these cells are phagocytic and defend the body against viruses and bacteria.
- These cells account for 3-9% of all leukocytes.



Agranulocytes

- **The Lymphocyte:**
- An agranular cell with very large nucleus for the size of the cell. This cell is much smaller than the three granulocytes. The T-lymphocytes act against virus infected cells and tumor cells. The B-lymphocytes leave the bloodstream to become plasma cells that produce antibodies.
- This is the second most numerous leukocyte, accounting for 25-35% of the cells counted in a Differential WBC Count.

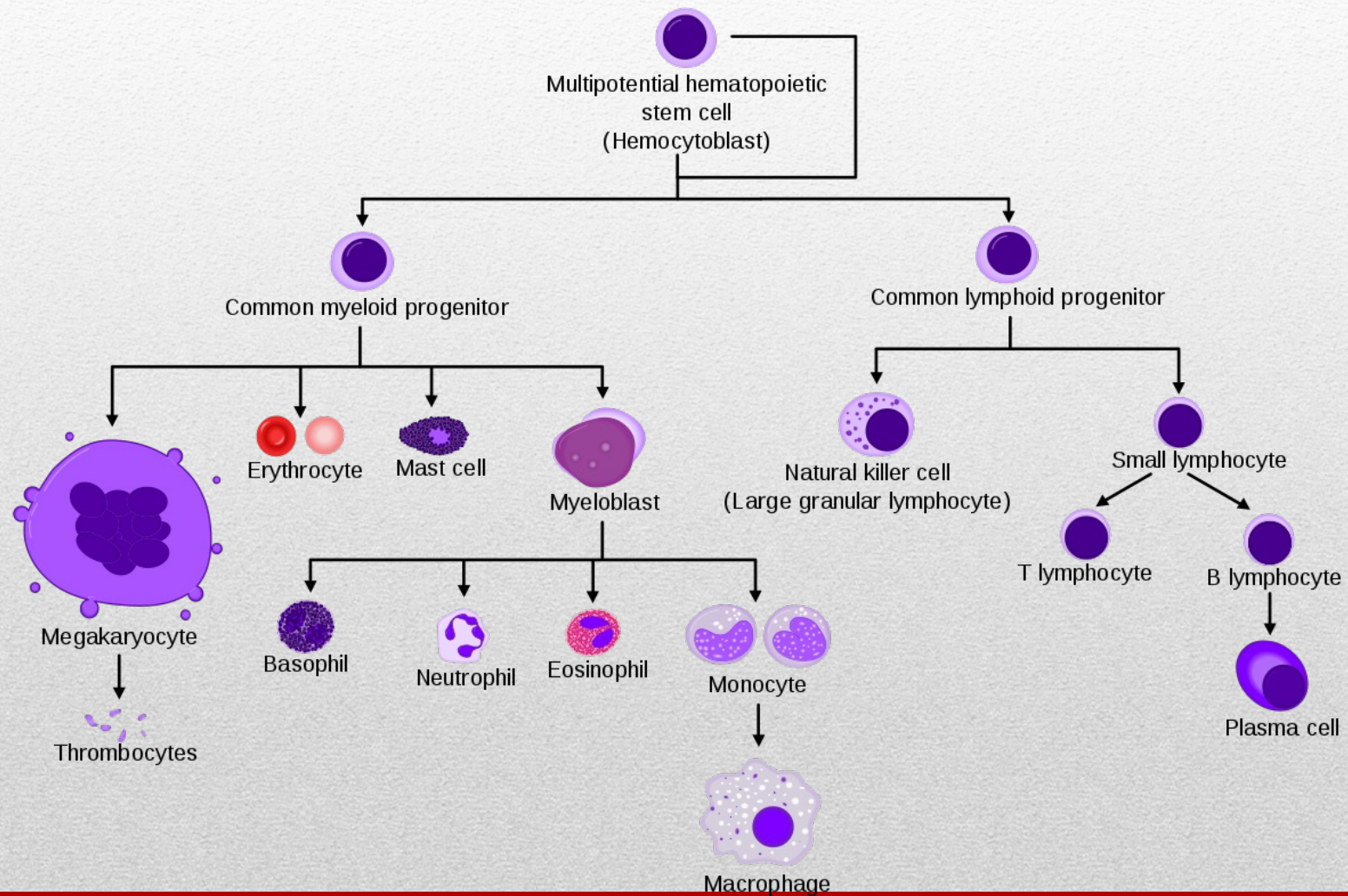


The Cellular Components - Derivative Cells

- **The Plasma Cell:**
 - Derived from B-lymphocytes. They produce antibodies.
- **The Histiocyte:**
 - a form of macrophage derived from extravascular monocytes



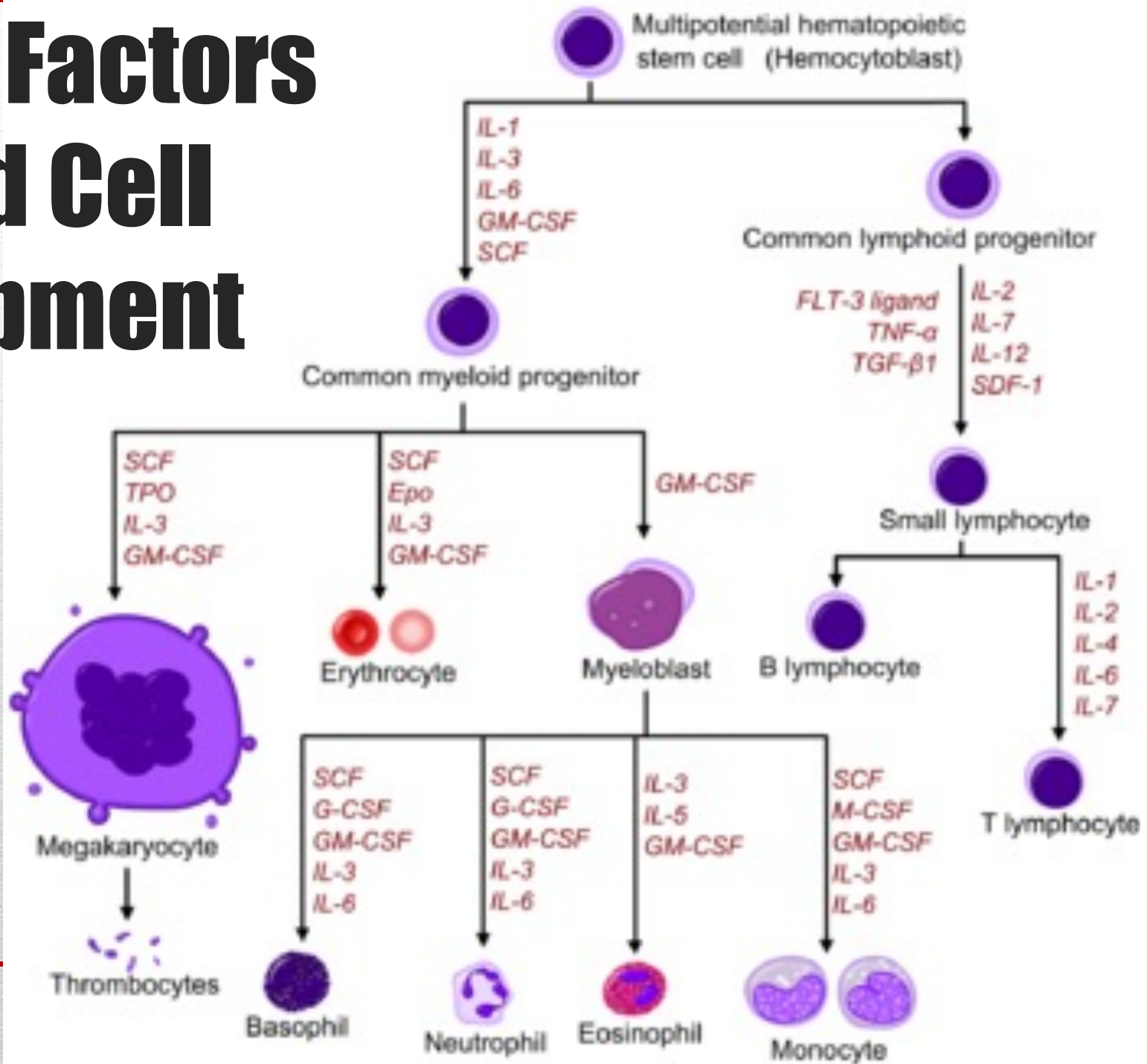
Hematopoiesis



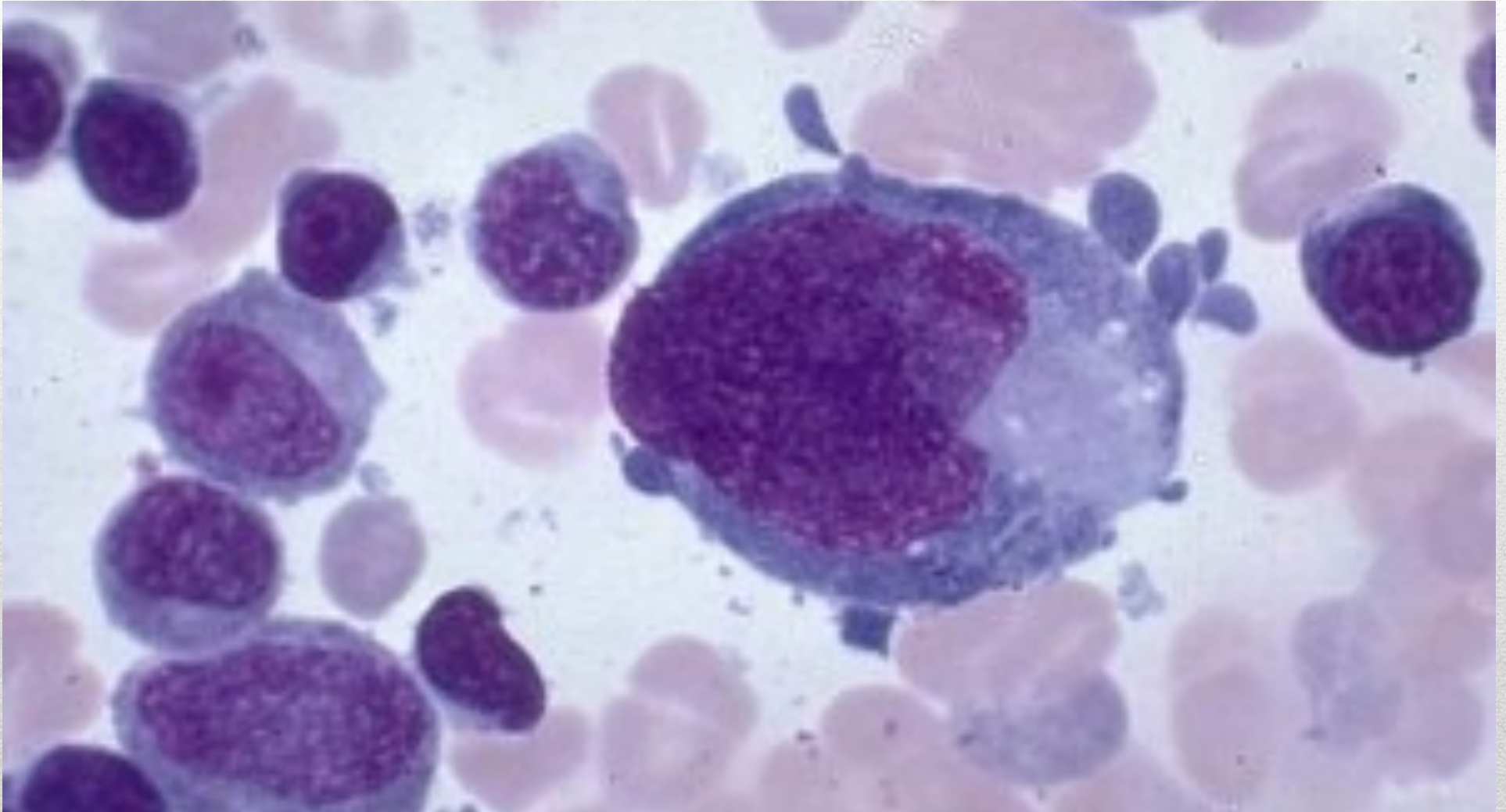
Hematopoiesis

- During embryogenesis this process occurs within islands of cells in the yolk sac
 - Post-partum this process begins in the bone marrow
 - But maturation of certain cells occurs in peripheral lymphoid organs (e.g. T cells in Thymus; B cells (bursa of Fabricius – in humans, the bone marrow))
 - Differentiation occurs under tight regulation via an array of interleukins, colony stimulating factors, TNF- α and TGF- β
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Growth Factors in Blood Cell Development



Development of Platelets



Platelets never have a nucleus. They are formed by pinching off a portion of cell membrane and cytoplasm (including granules) from the surface of a megakaryocyte.

- A glycoprotein hormone functioning as a cytokine acting on red cell precursors in the bone marrow.
- It is produced by interstitial fibroblasts in the kidney and peri-sinusoidal cells in the liver and its production and release is stimulated by hypoxia.
- Recombinant forms of EPO are available for treatment of anemia as following blood loss in inflammatory bowel disease, or after chemo or renal dialysis.

Erythropoietin - EPO
